

Chapter 17: Subspaces

Often a “big” vector space will have a (or many!) “little” vector space(s) inside of it.

Definition: Let V be a vector space and W be a nonempty subset of V (that is, $W \subseteq V$ with $W \neq \emptyset$). If W is a vector space with respect to the operations in V , then W is called a **subspace** of V .

Note: If $W \subseteq V$, then W automatically satisfies (A2), (A3), and (S2) – (S5), because

Furthermore, if W is closed under addition and scalar multiplication, then W satisfies (A4) and (A5), because

The two operations are inherited from V .

Therefore, if we know V is a vector space, we only need to check closure of W . More formally ...

Theorem: (Subspace Test) Let V be a vector space, and let W be a subset of V . Then W is a subspace of V if and only if the following conditions hold:

1. W is nonempty (i.e. W contains at least one vector)
2. If $\mathbf{u}$ and $\mathbf{v}$ are any vectors in W , then $\mathbf{u} + \mathbf{v}$ is in W .
3. If c is any real number and $\mathbf{u}$ is any vector in W , then $c\mathbf{u}$ is in W .

Example: The set $U = \left\{ \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} : x_1 x_2 = 0 \right\}$ in $\mathbb{R}^2$ is NOT a subspace of $\mathbb{R}^2$.

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \in U, \text{ but } \begin{bmatrix} 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} \notin U.$$

NOT closed under addition!

Example: Let W be the set of all vectors in $\mathbb{R}^3$ of the form $\begin{bmatrix} r \\ s \\ r+2s \end{bmatrix}$, where $r, s \in \mathbb{R}$.

Is W a subspace of $\mathbb{R}^3$?

Subspace test: ① $\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \in W$.

$$\text{② } \forall \begin{bmatrix} r_1 \\ s_1 \\ r_1+2s_1 \end{bmatrix}, \begin{bmatrix} r_2 \\ s_2 \\ r_2+2s_2 \end{bmatrix} \in W, \\ \begin{bmatrix} r_1+r_2 \\ s_1+s_2 \\ r_1+r_2+2(s_1+s_2) \end{bmatrix} \in W.$$

$$\text{③ } \forall \begin{bmatrix} r \\ s \\ r+2s \end{bmatrix} \in W, \begin{bmatrix} cr \\ cs \\ cr+2cs \end{bmatrix} \in W.$$

So W is a subspace.

Warning: To show a set IS a subspace, you NEED to use variables. To show a set is NOT a subspace, you should provide explicit (with numbers) counterexamples.

Example: Which of the following subsets of M_{22} with the usual operations of matrix addition and scalar multiplication are subspaces?

(a) W_1 is the set of all 2×2 matrices of the form

$$\begin{bmatrix} a & b \\ c & 0 \end{bmatrix}$$

W_1 is a subspace, since it satisfies all 3 condn's in the subspace test.

(b) W_2 is the set of all 2×2 matrices of the form

$$\forall \begin{bmatrix} a & b \\ c & 1 \end{bmatrix} \in W_2, \quad \begin{bmatrix} a_1 & b_1 \\ c_1 & 1 \end{bmatrix}, \begin{bmatrix} a_2 & b_2 \\ c_2 & 1 \end{bmatrix} \in W_2, \quad \begin{bmatrix} a_1+a_2 & b_1+b_2 \\ c_1+c_2 & 2 \end{bmatrix} \notin W_2,$$

so addition is NOT closed.

Example: Let W be the set of 2×2 symmetric matrices. Is W a subspace of M_{22} ?

Yes. ①. $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \in W$

② $\begin{bmatrix} a_1 & c_1 \\ c_1 & b_1 \end{bmatrix} + \begin{bmatrix} a_2 & c_2 \\ c_2 & b_2 \end{bmatrix} = \begin{bmatrix} a_1+a_2 & c_1+c_2 \\ c_1+c_2 & b_1+b_2 \end{bmatrix} \in W$

③ $s \cdot \begin{bmatrix} a & c \\ c & b \end{bmatrix} = \begin{bmatrix} sa & sc \\ sc & sb \end{bmatrix} \in W.$

Null Space and Image Space

Definition: Let A be an $m \times n$ matrix.

The **null space** of A , $\text{null}A$ is the set of vectors $\mathbf{x}$ in $\mathbb{R}^n$ which are solutions to $A\mathbf{x} = \mathbf{0}$, i.e.

$$\text{null}A = \{\mathbf{x} \in \mathbb{R}^n \mid A\mathbf{x} = \mathbf{0}\}.$$

This is also called the **solution space** of the homogeneous system $A\mathbf{x} = \mathbf{0}$.

The **image space** of A , $\text{im}A$, is the set of vectors $\mathbf{y}$ in $\mathbb{R}^m$ such that $A\mathbf{x} = \mathbf{y}$ has a solution, i.e.

$$\text{im}A = \{A\mathbf{x} \mid \mathbf{x} \in \mathbb{R}^n\}.$$

Theorem: For any $m \times n$ matrix A , the null space of A is a subspace of $\mathbb{R}^n$, and the image space of A is a subspace of $\mathbb{R}^m$.

Proof (we'll do one of the proofs in class):

$$\textcircled{1} \quad A \cdot \vec{0} = \vec{0} \quad , \Rightarrow \vec{0} \in \text{Null}(A)$$

$$\textcircled{2} \quad \forall \vec{x}_1, \vec{x}_2 \in \text{Null}(A) \quad , \quad A \cdot (\vec{x}_1 + \vec{x}_2) = A\vec{x}_1 + A\vec{x}_2 = \vec{0}$$

So $\vec{x}_1 + \vec{x}_2 \in \text{Null}(A)$

$$\textcircled{3} \quad \forall c \in \mathbb{R} \quad , \quad \vec{x} \in \text{Null}(A) \quad , \quad A \cdot (c\vec{x}) = c \cdot A\vec{x} = \vec{0}$$

Linear Combinations and Spanning Sets So $c\vec{x} \in \text{Null}(A)$.

Definition: Let $\mathbf{v}_1, \dots, \mathbf{v}_n$ be vectors in a vector space V . A vector $\mathbf{v}$ in V is a **linear combination** of $\mathbf{v}_1, \dots, \mathbf{v}_n$ if

$$\mathbf{v} = a_1 \vec{v}_1 + \dots + a_n \cdot \vec{v}_n$$

for some $a_1, \dots, a_n \in \mathbb{R}$. The $a_1, \dots, a_n$ are called the **coefficients** of $\mathbf{v}_1, \dots, \mathbf{v}_n$. The set of all linear combinations of these vectors is called their **span**, and is denoted

$$\text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_n\} = \{a_1 \mathbf{v}_1 + a_2 \mathbf{v}_2 + \dots + a_n \mathbf{v}_n \mid a_i \in \mathbb{R}\}$$

If every vector in V is a linear combination of the vectors in $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ (i.e. $V = \text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$), then these vectors are called a **spanning set** for V .

Example: Observe that every vector in $W = \left\{ \begin{bmatrix} r \\ s \\ r+2s \end{bmatrix} \mid r, s \in \mathbb{R} \right\}$ can be written as a linear combination of two (explicit) vectors:

$$\begin{aligned} \begin{bmatrix} r \\ s \\ r+2s \end{bmatrix} &= \begin{bmatrix} r \\ 0 \\ r \end{bmatrix} + \begin{bmatrix} 0 \\ s \\ 2s \end{bmatrix} \\ &= r \cdot \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} + s \cdot \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix} . \end{aligned}$$

Example: Consider $A = \begin{bmatrix} 5 & 1 \\ -1 & 1 \end{bmatrix}$ in M_{22} . Does A lie in

$\text{span} \left\{ \begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} -1 & 1 \\ 1 & 2 \end{bmatrix}, \begin{bmatrix} 0 & 2 \\ -2 & 1 \end{bmatrix} \right\}?$ Yes.

$$\begin{cases} 2x - y = 5 \\ y + 2z = 1 \\ x + y - 2z = -1 \\ x + 2y + z = 1 \end{cases} \Rightarrow \left[\begin{array}{ccc|c} 2 & -1 & 0 & 5 \\ 0 & 1 & 2 & 1 \\ 1 & 1 & -2 & -1 \\ 1 & 2 & 1 & 1 \end{array} \right]$$

$$\begin{array}{l} r_1 \leftrightarrow r_3 \\ \rightarrow \end{array} \left[\begin{array}{ccc|c} 1 & 1 & -2 & -1 \\ 0 & 1 & 2 & 1 \\ 2 & -1 & 0 & 5 \\ 1 & 2 & 1 & 1 \end{array} \right] \begin{array}{l} r_3 \rightarrow r_3 - 2r_1 \\ r_4 \rightarrow r_4 - r_1 \\ r_3 \rightarrow r_3 + 3r_2 \\ r_3 \rightarrow 1/10 r_3 \end{array} \left[\begin{array}{ccc|c} 1 & 1 & -2 & -1 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

Example: Let $\mathbf{v}_1 = x^2 + 2x + 1$ and $\mathbf{v}_2 = x^2 + 2$. Does $x^2 + 2x + 3$ lie in the span of $\{\mathbf{v}_1, \mathbf{v}_2\}$?

$$\begin{cases} a + b = 1 \\ 2a = 2 \\ a + 2b = 3 \end{cases} \text{ has NO sol'n!}$$

So $x^2 + 2x + 3$ does NOT lie in $\langle \vec{v}_1, \vec{v}_2 \rangle$

Example: $P_3 = \text{span}\{1, x, x^2, x^3\}$.

In fact, for any P_d , $P_d = \text{span}\{1, x, x^2, \dots, x^d\}$.

Example: Find a spanning set for $M_{2 \times 2}$.

$$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}.$$

Example: Find a spanning set for the null space of $A = \begin{bmatrix} 1 & -2 & -4 & 3 \\ 0 & -1 & -3 & 2 \\ 2 & 1 & 7 & -4 \end{bmatrix}$.

$$\begin{bmatrix} 1 & -2 & -4 & 3 \\ 0 & -1 & -3 & 2 \\ 2 & 1 & 7 & -4 \end{bmatrix} \xrightarrow[r_2 \rightarrow -r_2]{r_3 \rightarrow r_3 - 2r_1} \begin{bmatrix} 1 & -2 & -4 & 3 \\ 0 & 1 & 3 & -2 \\ 0 & 5 & 15 & -10 \end{bmatrix} \xrightarrow{r_3 \rightarrow r_3 - 5r_2}$$

$$\begin{bmatrix} \textcircled{1} & -2 & -4 & 3 \\ 0 & \textcircled{1} & 3 & -2 \\ 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{r_1 \rightarrow r_1 + 2r_2} \begin{bmatrix} 1 & 0 & 2 & -1 \\ 0 & 1 & 3 & -2 \\ 0 & 0 & 0 & 0 \end{bmatrix} \Rightarrow \begin{cases} x_1 = -2s + t \\ x_2 = -3s + 2t \\ x_3 = s \\ x_4 = t \end{cases}$$

$$\Rightarrow \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = s \cdot \begin{bmatrix} -2 \\ -3 \\ 1 \\ 0 \end{bmatrix} + t \cdot \begin{bmatrix} 1 \\ 2 \\ 0 \\ 1 \end{bmatrix}.$$

Example: Let $\mathbf{u}, \mathbf{v}$ be two vectors in a vector space V . Show that $\text{span}\{\mathbf{u}, \mathbf{v}\}$ is a subspace of V .

$$\textcircled{1} \quad 0 \cdot \vec{u} + 0 \cdot \vec{v} = \vec{0} \in \text{span}\{\vec{u}, \vec{v}\}$$

$$\textcircled{2} \quad \vec{u} + \vec{v} \in \text{span}\{\vec{u}, \vec{v}\}$$

$$\textcircled{3} \quad c \cdot \vec{u} \in \text{span}\{\vec{u}, \vec{v}\}.$$

Theorem: Let $U = \text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ in a vector space V . Then:

1. U is a subspace of V containing each of $\mathbf{v}_1, \dots, \mathbf{v}_n$.
2. U is the “smallest” subspace containing these vectors in the sense that any subspace that contains each of $\mathbf{v}_1, \dots, \mathbf{v}_n$ must contain all of U .

Fact: Given vectors $\mathbf{v}_1, \dots, \mathbf{v}_k$ in a vector space V and a subspace U of V ,

$$\text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_k\} \subseteq U \text{ if and only if each } \mathbf{v}_i \text{ is an element of } U.$$

Theorem: Suppose S and S' are two sets of vectors in a vector space.

- a. If each vector in S can be expressed as a linear combination of the vectors in S' , then $\text{span} S$ is a subspace of $\text{span} S'$.
- b. If each vector in S can be expressed as a linear combination of the vectors in S' , and each vector in S' can be expressed as a linear combination of the vectors in S , then $\text{span} S$ and $\text{span} S'$ are the same subspace.

Example: Show that $P_2 = \text{span}\{2 - 2x^2, 4x + 2, 7x^2\}$.

It is clear that

$$A = \text{span}\{2 - 2x^2, 4x + 2, 7x^2\} \subseteq \text{span}\{1, x, x^2\} = P_2$$

"B"

$$\text{Since } 1 = \frac{1}{2} \times (2 - 2x^2) + \frac{1}{7} \times 7x^2$$

$$x = -\frac{1}{4} (2 - 2x^2) + \frac{1}{4} (4x + 2) - \frac{1}{14} \times 7x^2$$

$$x^2 = \frac{1}{7} \cdot 7x^2$$

$$\Rightarrow B \subseteq A. \quad \text{So } A = B.$$